

Let $\sum_{i=1}^n x_i = 1$ for $x_1, \dots, x_n > 0$. Show that the following inequality holds.

$$\left(\sum_{i=1}^n x_i^2\right)^2 \sqrt{n} \geq \sum_{i=1}^n x_i^3$$

$$\left(\sum_{i=1}^n x_i^2\right)^4 n \geq \left(\sum_{i=1}^n x_i^3\right)^2 \quad \begin{matrix} a^2 \geq b^2 \\ |a| \geq |b| \end{matrix}$$

$$\left(\sum_{i=1}^n x_i^2\right)^3 \underbrace{\left(\sum_{i=1}^n x_i^2\right) \left(\sum_{i=1}^n 1\right)}_{a \geq b} \geq \left(\sum_{i=1}^n x_i^2\right)^3 \left(\sum_{i=1}^n x_i\right)^2$$

$$\left(\sum_{i=1}^n x_i^2\right)^3 \geq \frac{\left(\sum_{i=1}^n x_i^3\right)^2}{\left(\sum_{i=1}^n x_i^2\right)^3} \quad n \geq 1$$

$$\underbrace{\left(\sum_{i=1}^n x_i^3\right)^2}_{\leq} \leq \underbrace{\left(\sum_{i=1}^n x_i^2\right) \left(\sum_{i=1}^n x_i^4\right)}_{= \left(\sum_{i=1}^n x_i^2\right)^3} \leq \left(\sum_{i=1}^n x_i^2\right) \left(\sum_{i=1}^n x_i^2\right)^2$$

□